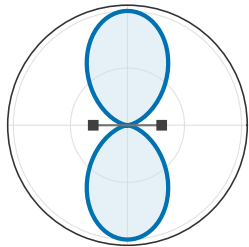
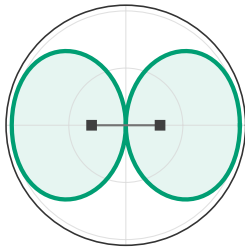


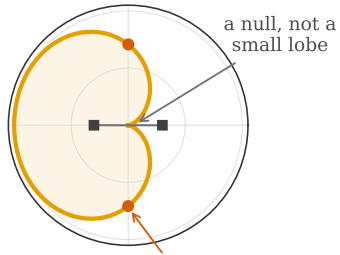
$d = \lambda/2, \phi = 0$
figure-eight,
broadside



$d = \lambda/2, \phi = 180^\circ$
figure-eight,
end-fire



$d = \lambda/4, \phi = 90^\circ$
cardioid



—3 dB points, so the
beamwidth is 180°